17_multiple_imputation_linar_regression

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```
#In this script
#0. Use multiply imputed datasets created in 2 multiple imputation
#1. Run linear mixed models analysis for eight outcome vars as specified in pre-registration
#2. Create tables and figures for manuscript and supplemental material

rm(list = ls())

#Load packages
library(pacman)
p_load(
  dplyr,
  tidyr,
  broom.mixed,
  lme4,
  ggplot2,
  mitml,
  mice,
  emmeans,
  ordinal,
  ggpubr,
  ordinalimputation,
  magrittr,
  see,
  patchwork,
  boot,
  lmeresampler
)
```

```
## Installing package into 'C:/Users/anne.gaule/AppData/Local/R/win-library/4.6'
## (as 'lib' is unspecified)

## Warning: package 'ordinalimputation' is not available for this version of R
##
## A version of this package for your version of R might be available elsewhere,
## see the ideas at
## https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#Installing-packages

## Warning: unable to access index for repository http://www.stats.ox.ac.uk/pub/RWin/bin/windows/contri
##   cannot open URL 'http://www.stats.ox.ac.uk/pub/RWin/bin/windows/contrib/4.6/PACKAGES'

## Warning: 'BiocManager' not available.  Could not check Bioconductor.
```

```
##
## Please use ‘install.packages('BiocManager')‘ and then retry.

## Warning in p_install(package, character.only = TRUE, ...):

## Warning in library(package, lib.loc = lib.loc, character.only = TRUE,
## logical.return = TRUE, : there is no package called 'ordinalimputation'

## Warning in p_load(dplyr, tidyr, broom.mixed, lme4, ggplot2, mitml, mice, : Failed to install/load:
## ordinalimputation
```

```
#Set working directory
file_path_read_data_diaries <- 'X:/Viding/Prosocial Project/7 Analyses/8_Diaries/1_Data_Input/'

#Load the imputed data
#imputed_full <- readRDS(paste0(file_path_read_data_diaries, "imputed_full_35.rds")) ## this is the mod
imputed_full <- readRDS(paste0(file_path_read_data_diaries, "imputed_reviewer.rds")) ## this is the mod
```

1. PRIMARY: Child antisocial behaviour T2

Fit model:

```
## child antisocial behaivour
fit <- with(imputed_full, lm(T2_child_A ~ Training_Condition + T1_child_A ))
pooled <- pool(fit)
```

Examine results:

```
summary(pooled, conf.int = TRUE)
```

```
##                       term   estimate  std.error  statistic        df
## 1              (Intercept)  6.4578158 1.95847081  3.2973766 116.40688
## 2 Training_ConditionTRAIN -1.2001791 2.32941970 -0.5152267  99.81136
## 3              T1_child_A  0.6658754 0.05407115 12.3148010 110.91774
##        p.value      2.5 %     97.5 %   conf.low  conf.high
## 1 1.294563e-03  2.5789604 10.3366711  2.5789604 10.3366711
## 2 6.075339e-01 -5.8217884  3.4214302 -5.8217884  3.4214302
## 3 1.735977e-22  0.5587289  0.7730219  0.5587289  0.7730219
```

Check FMI:

```
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                       term       fmi
## 1              (Intercept) 0.1744631
## 2 Training_ConditionTRAIN 0.2418179
## 3              T1_child_A 0.1963973
```

2. PRIMARY: Teacher antisocial behaviour

Fit model:

```
## teacher antisocial behaivour
fit <- with(imputed_full, lm(T2_teach_A ~ Training_Condition + T1_teach_A ))
pooled <- pool(fit)
```

Examine results:

```
summary(pooled, conf.int = TRUE)
```

```
##                         term   estimate  std.error statistic       df      p.value
## 1                (Intercept) 10.4743231 1.87694797 5.5805080 128.6020 1.358422e-07
## 2 Training_ConditionTRAIN  1.6982641 2.23637714 0.7593818 123.9846 4.490656e-01
## 3               T1_teach_A  0.3658305 0.06074877 6.0220236 138.2440 1.471482e-08
##        2.5 %     97.5 %   conf.low  conf.high
## 1  6.7606267 14.1880194  6.7606267 14.1880194
## 2 -2.7281581  6.1246863 -2.7281581  6.1246863
## 3  0.2457136  0.4859474  0.2457136  0.4859474
```

Check FMI:

```
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                         term        fmi
## 1                (Intercept) 0.12550369
## 2 Training_ConditionTRAIN 0.14420479
## 3               T1_teach_A 0.08460089
```

SECONDARY: Prosocial T2

1. PRIMARY: Child prosocial behaviour T2

Fit model:

```
## child antisocial behaivour
fit <- with(imputed_full, lm(T2_child_P ~ Training_Condition + T1_child_P ))
pooled <- pool(fit)
```

Examine results:

```
summary(pooled, conf.int = TRUE)
```

```
##                         term  estimate std.error statistic        df      p.value
## 1                (Intercept) 4.0147375 3.9322055  1.020989  90.71965 3.099748e-01
## 2 Training_ConditionTRAIN 3.8742308 2.7314489  1.418379 103.07669 1.590964e-01
## 3               T1_child_P 0.9011837 0.0662454 13.603717 107.76670 3.888720e-25
##        2.5 %    97.5 %   conf.low conf.high
## 1 -3.7964309 11.825906 -3.7964309 11.825906
## 2 -1.5429060  9.291368 -1.5429060  9.291368
## 3  0.7698706  1.032497  0.7698706  1.032497
```

Check FMI:

```r
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                       term       fmi
## 1              (Intercept) 0.2809962
## 2 Training_ConditionTRAIN 0.2282494
## 3              T1_child_P 0.2090973
```

2. SECONDARY: Teacher prosocial behaviour T2

Fit model:

```r
## teacher antisocial behaivour
fit <- with(imputed_full, lm(T2_teach_P ~ Training_Condition + T1_teach_P ))
pooled <- pool(fit)
```

Examine results:

```r
summary(pooled, conf.int = TRUE)
```

```
##                       term   estimate  std.error statistic       df      p.value
## 1              (Intercept) 14.5232017 2.77906003  5.225940 121.4243 7.300022e-07
## 2 Training_ConditionTRAIN  4.5115445 2.83878317  1.589253 127.3068 1.144837e-01
## 3              T1_teach_P  0.5665591 0.06945501  8.157210 124.5913 3.154681e-13
##        2.5 %     97.5 %   conf.low  conf.high
## 1  9.0215136 20.0248899  9.0215136 20.0248899
## 2 -1.1057649 10.1288539 -1.1057649 10.1288539
## 3  0.4290946  0.7040236  0.4290946  0.7040236
```

Check FMI:

```r
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                       term       fmi
## 1              (Intercept) 0.1544603
## 2 Training_ConditionTRAIN 0.1307852
## 3              T1_teach_P 0.1417657
```

Secondary : T3

1. PRIMARY: Child antisocial behaviour T3

Fit model:

```r
## child antisocial behaivour
fit <- with(imputed_full, lm(T3_child_A ~ Training_Condition + T1_child_A ))
pooled <- pool(fit)
```

Examine results:

```r
summary(pooled, conf.int = TRUE)
```

```
##                         term   estimate  std.error  statistic       df
## 1                (Intercept)  7.0447848 2.26025686  3.1168072 103.3261
## 2 Training_ConditionTRAIN -0.5147476 2.56512544 -0.2006715 105.4654
## 3                T1_child_A  0.5891642 0.06035669  9.7613731 111.4422
##        p.value      2.5 %     97.5 %   conf.low  conf.high
## 1 2.367743e-03  2.5622666 11.5273031  2.5622666 11.5273031
## 2 8.413419e-01 -5.6006559  4.5711606 -5.6006559  4.5711606
## 3 1.226549e-16  0.4695686  0.7087598  0.4695686  0.7087598
```

Check FMI:

```r
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                         term       fmi
## 1                (Intercept) 0.2272218
## 2 Training_ConditionTRAIN 0.2184514
## 3                T1_child_A 0.1942932
```

2. PRIMARY: Teacher antisocial behaviour

Fit model:

```r
## teacher antisocial behaivour
fit <- with(imputed_full, lm(T3_teach_A ~ Training_Condition + T1_teach_A ))
pooled <- pool(fit)
```

Examine results:

```r
summary(pooled, conf.int = TRUE)
```

```
##                         term  estimate  std.error statistic       df      p.value
## 1                (Intercept) 9.1058045 1.90523175  4.779368 126.9314 4.781041e-06
## 2 Training_ConditionTRAIN 7.0384537 2.30981621  3.047192 114.8397 2.866525e-03
## 3                T1_teach_A 0.3032959 0.06209529  4.884362 133.6100 2.917567e-06
##      2.5 %     97.5 % conf.low  conf.high
## 1 5.335675 12.8759339 5.335675 12.8759339
## 2 2.463084 11.6138230 2.463084 11.6138230
## 3 0.180479  0.4261128 0.180479  0.4261128
```

Check FMI:

```r
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                         term       fmi
## 1                (Intercept) 0.1323101
## 2 Training_ConditionTRAIN 0.1807105
## 3                T1_teach_A 0.1046899
```

1. PRIMARY: Child prosocial behaviour T3

Fit model:

```
## child antisocial behaivour
fit <- with(imputed_full, lm(T3_child_P ~ Training_Condition + T1_child_P ))
pooled <- pool(fit)
```

Examine results:

```
summary(pooled, conf.int = TRUE)
```

```
##                       term  estimate  std.error statistic       df      p.value
## 1              (Intercept) 9.1765252 4.25237226  2.157978 108.6885 3.313074e-02
## 2 Training_ConditionTRAIN 4.0479528 3.08839813  1.310697 104.5535 1.928321e-01
## 3              T1_child_P 0.7571348 0.07380724 10.258274 115.4959 6.062095e-18
##        2.5 %     97.5 %   conf.low  conf.high
## 1  0.7481905 17.6048599  0.7481905 17.6048599
## 2 -2.0760750 10.1719806 -2.0760750 10.1719806
## 3  0.6109436  0.9033261  0.6109436  0.9033261
```

Check FMI:

```
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                       term       fmi
## 1              (Intercept) 0.2053705
## 2 Training_ConditionTRAIN 0.2221804
## 3              T1_child_P 0.1780936
```

2. SECONDARY: Teacher prosocial behaviour

Fit model:

```
## teacher antisocial behaivour
fit <- with(imputed_full, lm(T3_teach_P ~ Training_Condition + T1_teach_P ))
pooled <- pool(fit)
```

Examine results:

```
summary(pooled, conf.int = TRUE)
```

```
##                       term   estimate  std.error statistic       df      p.value
## 1              (Intercept) 18.7535574 2.89564464  6.476471 128.6568 1.811010e-09
## 2 Training_ConditionTRAIN -3.8369753 2.90592863 -1.320396 141.9879 1.888274e-01
## 3              T1_teach_P  0.3708975 0.07004447  5.295171 145.3958 4.305003e-07
##        2.5 %     97.5 %   conf.low  conf.high
## 1 13.0243089 24.4828060 13.0243089 24.4828060
## 2 -9.5814512  1.9075006 -9.5814512  1.9075006
## 3  0.2324606  0.5093344  0.2324606  0.5093344
```

Check FMI:

```
pool(fit)$pooled[, c("term", "fmi")]
```

```
##                        term        fmi
## 1               (Intercept) 0.12527940
## 2 Training_ConditionTRAIN 0.06745627
## 3               T1_teach_P 0.05077528
```